

7SSMM616 Option Analysis & Valuation

Coursework

30% of final grade of module

Submission deadline: **September 30**, 2026 at 11:00am

The indicative maximum word length, excluding appendices containing spreadsheet results, is **2,000 words**. The submission deadline is **30 September, 2026**, at 23:50. This coursework is an individual assignment. Each student must submit their own coursework. In the cover page of your coursework report, please specify your ID number.

The coursework should be submitted electronically on the Keats 616 platform (a submission link will appear on Keats near the deadline). You **are required** to submit your excel or other supporting application files. **These files should not exceed 500MB**. If it is technically impossible to upload all your files on KEATS, you can use any kind of cloud-based services (i.e., Dropbox, Google Cloud, WeTransfer, Microsoft Azure, and others) to upload your files. In this case provide (write) the web link in the first page of your coursework. This coursework carries 30% of the assessed marks for this module.

Solve questions 1 and 2 using binomial trees with 20 time steps. Solve question 3 using Monte Carlo Simulations. Provide all necessary details on parameters, formulas, and methodology applied in each question. **The clarity of presentation will be evaluated as 30% of the total mark, encompassing organization, limitations analysis, originality, and applicability of theoretical concepts.**

1. Estimate the value of European and American put options on Tesla, Inc. (TSLA) stock. Use real stock price closed at 31 July 2026 and equal strike price. Use an appropriate US risk-free rate. The time to maturity is 6 (six) months. Use daily stock prices (of Tesla) to calculate the volatility and any other required information. Discuss your findings.

[mark 25%]

2. A compound option is an option for which the underlying asset is another option (option on option). Therefore, there are two strike prices and two exercise dates. The strike price X_2 of the second option is \$20 (the strike price of the first option remains the same as question 1), and the time to maturity of first option T_1 is 1 year and for the second option T_2 equals 2 years. Consider both options as European Call options. Estimate the price of the compound option if the underlying asset of the second option is Tesla. Use all relevant information for Tesla as in question 1.

[mark 35%]

3. Estimate the value of the following options using Monte Carlo simulations with 20 time steps and 1,000 simulations. The use of Antithetic Variate technique for variance reduction will earn additional marks.

Originality: Use your own estimations and/or programming to find the following prices.

α . European call and put options on Tesla, Inc. (TSLA) stock with the same real parameters for stock price, strike price, volatility, risk free rate and time to maturity as in question 1.

β . Apply the methodology of Longstaff and Schwartz (2001) for the pricing of American options using Monte Carlo Simulations to find the price of the American call and put options on Tesla stock. Use the same parameters as in (3a).

[mark 40%]

References

- Broadie M., P. Glasserman and G. Jain (1997), "Enhanced Monte Carlo Estimates for American Option prices", *Journal of Derivatives* 5: 25-44.
- Boyle Phelim, Broadie Mark, Glasserman (1997), "Monte Carlo Methods for Security Pricing", *Journal of Economic Dynamics and Control* 21: 1267-1321.
- Boyle P. Phelim (1977), "Options: A Monte Carlo Approach", *Journal of Financial Economics* 4: 323-338.
- Longstaff A. Francis and Schwartz S. Eduardo (2001), "Valuing American options by Simulation: A Simple Least-Squares Approach", *Review of Financial Studies* 14: 113-147.